

Stuck behind the keyboard

Years of keyboarding and mousing later, it finally looks like we'll be breaking our shackles and poking at computers in new and wonderful ways. Famous last words?

Ahmed Shaikh

We shake our tiny fists in outrage at the fact that we are still using the mouse and keyboard to interact with our silly computing machines. Join us in our demand for a revolution that is way overdue.

So here we are, mid-way through 2008. I'm currently typing on a keyboard based on a patent filed by Christopher Latham Sholes' in 1868. That's 140 years ago! Being generous, let's count down from the 1970s when keyboards very not very different from what we use today—that's still a healthy 38 years of being stuck hitting keys.

Where is the revolution? Why can't I download the latest episode of *Battlestar Galactica* by just thinking about it during a boring Monday meeting? Why do we have to struggle with interfaces, to this late date, when our machines should be bending time and space to make things easier for us instead!

In many ways, input devices seem to have taken a step back. This might be subjective, but does today's membrane keyboard really match up to the visceral feedback that the not-so-old mechanical keys brought to bear? How about the myriad keys surrounding the

QWERTY layout, threatening to throw themselves at your fingertips to inadvertently launch a calculator application, or start an instant messenger, or much worse—put your computer to sleep! Who asked for those anyway? In 38 years, all we have been offered is a wireless setup, more keys to mistakenly hit, and LCD panels that tell us whether it's hot outside, or if you have unread mail.



Shrikishna Patkar

It's all pretty glum, if you ask us. The mouse does not fare any better. Where we are today, is safely stuck with the status quo. No one seems to have either the courage to introduce a new interface, or the vision to dream up a newer manner to communicate with our machine brethren. Or so it would seem.

Fact is that our being stuck with the same interface for decades is not entirely a matter of manufacturer laziness or consumer ennui. Human-machine interface is an actively researched field. New means of interaction with the computer are constantly being looked into, and either pursued or discarded as being impractical. Consumers too, are very open to exploring options other than the keyboard and mouse. The acceptance of the Apple iPhone, the Nintendo DS and the Wii console stands evidence to this.

So, where is the revolution?

Touch The Future

When the Nintendo DS was introduced, it made mainstream what was until then for a few PDA-philes—the touch screen. In 1971, the first step in touch-screen technology was taken by Dr Sam Hurst, while he was an instructor at the University of Kentucky. Three years later, around 1974, he created a sensor called the Elograph, which was later developed into a transparent surface, similar to what we use today. Fast forward to 1993 and Apple introduces the Newton—a touch-screen driven, monochromatic device dubbed the Personal Digital Assistant. But success only came to touch-screens with the 1996 arrival of the Palm Pilot. The touch-screens used by these devices were built to take in a single point of input. Touch-screens are generally based on resistive technologies, wherein the pressure from a stylus or your finger causes conductive and resistive layers of circuitry to touch each other, changing the circuit's resistance. These screens would only take in one input at a time—generally the first point of contact was considered and if you so happened to touch the screen with two fingers, the second point of contact would be ignored completely.

That changed with the introduction of the iPhone in 2007—multi-touch is the new buzzword, and has enough traction to trickle into devices other than the iPhone and iPod Touch. Apple's implementation of multi-touch is based on capacitive circuitry beneath the screen. Due to this, the touch screen can only be driven by fingers and not the stylus. What makes multi-touch impressive, is the interfaces it



The iPhone has changed the way we interact with our devices by bringing us gesture-based commands

provides—touch to start a program, use gestures to interact—pinching zooms out of a screen, swipe your finger to move images, and so on. It's immediately intuitive—the hallmark of a good interface.

The success of this interface setup has spawned many me-too products ranging from the poorly implemented (the HTC Touch), to more recent implementations which seem to do better such as the touch screen used in the HTC Diamond, or the scroll pad used in the Eee PC 900 laptop.

While multi-touch screens and surfaces have, thus far, seen commercial use in smaller devices, there are larger plans ahead.

Microsoft Surface, for example, was showcased not too long ago. It successfully captured the imagination of all of

us bored by the humdrum and tedious keyboard and mouse interface. The Surface was announced on May 29, 2007. The particular example shown was that of a large table, the 'surface' of which, was covered by a multi-touch interface. It is a hardware and software soup that allows one or more users to manipulate digital content using multi-touch interactions—natural motions, hand gestures, and even physical objects to share and exchange data intuitively. Beneath the surface is a Microsoft Vista PC, running a 30-inch reflective screen. A projector beneath the screen throws the interface elements visible on the Surface, while five cameras track and record reflections of infrared light from a person's fingertips. Electronic devices can also be tagged and used directly on the Surface. The cameras recognise these tagged devices and allow meaningful interactions. For example, placing a cell phone on the Surface would 'spill' all the photographs inside the camera onto the Surface. The user could then interact with these photographs—write on them, resize them using hand gestures, delete or rotate them, and even move them to other devices by flipping through them with a finger.

The technology beneath the Surface makes it inherently costly for wide-spread use. For now, Microsoft will ship the Surface to commercial clients such as AT&T and T-Mobile, since each Surface is estimated around \$5,000 to \$15,000! However, Microsoft expects prices to fall to consumer levels by the year 2010.

While Surface seems a few years outside our grasp, another technology, again from Microsoft, promises an early entry into our lives.

Dubbed the TouchWall, this implementation actually consists of two pieces—the hardware itself